Andreas Tzitzikas

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Education

University of Maryland, College Park, MD — Expected May 2027
Master of Science in Computer Engineering — **GPA: 4.00**
Bachelor of Science in Computer Engineering (GPA: 3.88) — Expected May 2026
Relevant Coursework: Digital Computer Design (ENEE646), Digital CMOS VLSI Design (ENEE640), Microprocessors (ENEE440), Computer Organization (ENEE350), Digital Circuits Lab (ENEE245), PCB Design (ENEE459A)

Projects

PCB Design with RP2350 Microcontroller | KiCad, RP2350 Firmware — Jan 2024 – Present

- Designed complete PCB system integrating RP2350 microcontroller with QFN60 package, implementing proper power distribution, signal integrity, and thermal management for reliable operation.
- Developed embedded firmware for sensor interfaces and peripheral control, ensuring robust communication protocols and real-time performance requirements.
- Generated fabrication-ready documentation including Gerber files, bill of materials, and assembly instructions for JLCPCB manufacturing with comprehensive design validation.

STM32G4 Bare-Metal Peripheral Integration | ARM Assembly, Embedded C — Jan 2025 – May 2025

- Implemented bare-metal firmware with direct register-level programming for GPIO, DAC, ADC, TIM2 (PWM/PFM), and SPI2 flash memory drivers on STM32G4 microcontroller.
- Engineered shared interrupt system multiplexing single EXTI1 hardware interrupt to four distinct peripheral drivers using state-based arbitration for efficient resource utilization.
- Developed comprehensive testing framework ensuring real-time performance, peripheral synchronization, and robust error handling in resource-constrained embedded environment.

Tomasulo Out-of-Order RISC-V CPU | RTL Design, FPGA Implementation, Verification — Nov 2025 - Dec 2025

- Architected complex RTL system using Tomasulo's Algorithm with register renaming, 8-entry reservation stations, and 16-entry reorder buffer for advanced hardware design.
- Achieved 36% higher operating frequency (69.7MHz vs 51.2MHz) and 40-150% IPC improvement compared to in-order implementation through hardware optimization and performance analysis.
- Implemented sophisticated branch prediction, speculative execution, and precise exception handling ensuring robust hardware behavior across complex computational workloads.
- Synthesized complex digital system with 1,652 standard cells achieving zero DRC violations and comprehensive functional verification for reliable hardware operation.

Polynomial Evaluation Accelerator | SystemVerilog, Digital Design — Fall 2025

- Architected dataflow-based accelerator with 5 finite state machines managing FIFO handshaking and resource sharing across 9 processing modules for high-throughput computation.
- Designed dual-port coefficient memory with auto-padding logic enabling efficient batch processing of polynomial evaluations with minimal memory access latency.
- Achieved 100% functional coverage through comprehensive testbench development with 15+ complex test cases validated against reference C implementation.

Experience

Embedded Systems Intern, Alchemity – College Park, MD — Jun 2025 – Aug 2025

- Accelerated validation of a hybrid solid oxide fuel cell reactor by developing a full-stack automation suite (**Rust**, **Python**) to parallelize testing, converting multi-day manual experiments into overnight runs to significantly increase throughput.
- Engineered real-time **STM32** firmware to precisely control material synthesis within modular reactors, featuring non-blocking motor/relay drivers and **SPI** peripheral management.
- Built end-to-end control interfaces including desktop GUIs, embedded display drivers, and CLI tools to streamline workflows between hardware and software teams.

Skills

Languages & HDLs: SystemVerilog, Verilog, C/C++, ARM Assembly, Python
Hardware Design: Digital Logic Design, PCB Design, Microcontroller Programming, FPGA Implementation, CMOS VLSI Design
Embedded Systems: STM32 Bare-Metal Firmware, Real-Time Control, SPI, I2C, UART, GPIO, ADC/DAC, PWM
EDA Tools: KiCad, Xilinx Vivado, Icarus Verilog, Verible, GTKWave, LPKF Prototyping, Altium Designer
Verification: Functional Verification, Testbench Development, Static Timing Analysis, Linux